Learning a Data-Driven Beam Search Heuristic for the Gapped Longest Common Subsequence Problem

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1 Introduction

The *Longest Common Subsequence Problem* (LCSP) [1] is a well-known combinatorial optimization problem with numerous applications in computational biology, particularly in the analysis of molecular sequences. Given an arbitrary set $S = \{s_1, \ldots, s_m\}$ of input sequences over an alphabet Σ, the goal is to determine the longest possible subsequence common to all $s_i \in S$. In recent decades, many practically motivated variants of the LCSP have been proposed by considering additional structural or biological constraints such as the constrained, arc-preserving, repetition-free version, etc. This work is focused on solving the *Variable Gapped* LCSP (VGLCSP), introduced in [3] and theoretically examined in [2]. This problem extends the standard LCSP by imposing flexible distance limits ("gaps") between consecutive symbols in the resulting subsequence, allowing these gaps to vary along the input sequences. Formally, for each input sequence $s_i \in S$, a gap function $G_{s_i} : \{1, \ldots, |s_i|\} \mapsto \mathbb{N}$ defines the maximum distance allowed between consecutive characters of a subsequence. That is, if the characters of a subsequence s appear at the positions $i_i^1 < \ldots < i_i^{|s|}$ in s_i, we say that the gap constraint is satisfied if $i_i^x - i_i^{x-1} \leq G_{s_i}[i_i^x] + 1$ for all $x = 2, \ldots, |s|$. A common subsequence s is considered feasible if the corresponding gap constraints hold across all sequences in S. The VGLCSP represents a flexible and realistic model of sequence alignment, which is particularly useful for comparing DNA or protein sequences with variable structural distances. Although exact dynamic programming approaches exist for the two-sequence case ($m = 2$), their computational complexity becomes prohibitive for larger m. To the best of our

knowledge, no effective exact or approximate approaches currently exist for the generalized VGLCSP, a clearly NP-hard problem.

Building on the well-established state-space representation of the LCSP [1], we propose a general *state graph* formulation for the VGLCSP. Each node represents a partial feasible subsequence, characterized by the vector of current positions in each input sequence and the subsequence length. Specifically, a state $v = (\mathbf{p}^{L,v}, l^v)$ is induced by a partial solution s^v if: (i) $p_i^{L,v}-1$ is the smallest index such that s^v is a subsequence of the prefix string $s_i[1]\cdots s_i[p_i^{L,v}-1]$; (ii) $l^v = |s^v|$; and (iii) all the gaps G_{s_i} are preserved with s^v. A directed arc $\alpha = (v_1 v_2)$, labeled $lett(\alpha) = a \in \Sigma$, exists between states v_1 and v_2 if $l^{v_2} = l^{v_1}+1$ and $s^{v_2} = s^{v_1}\cdot a$. To expand a state v, all letters that occur in each suffix of s_i defined by respective positions $p_i^{L,v}$ are considered. For each such letter a, the minimal positions $p_i^{L,a} \geq p_i^{L,v}$ of its appearances are identified fulfilling $p_i^{L,a} - p_i^{L,v} \leq G_{s_i}(p_i^{L,a})$. The resulting child node, if not suboptimal, is given by $w = (p^{L,a}+1, l^v+1)$. The root node $r := ((1,\ldots,1),0)$ represents the empty subsequence, while goal states correspond to non-extendable feasible subsequences. Unlike the standard LCSP, the VGLCSP admits multiple (possibly exponentially many) root nodes. For example, in sequences $s_1 =$ `ATGGAAAA` and $s_2 =$ `ATCCAAAA` with $G_{s_1} = G_{s_2} = 1$, the state $((1,1),0)$ cannot reach any state with the position vector $(5,5)$ via transitions, missing optimal subsequence `AAA`. This motivates the methodology of *Iterative Multi-Source Beam Search* (IMSBS). BS is a popular meta-heuristic approach that performs in a breadth-first-search manner, controlled by the size of beam β and a heuristic h. In the IMSBS, BS is iteratively executed with a set of promising root nodes as an initial beam, selected from a dynamically maintained global pool of root nodes, controlling diversification of the approach. Each iteration performs a complete BS, guided by a heuristic h based on letter frequencies, minimal residual substring lengths, or probability-weighted estimates.

1.1 Contributions

This work proposes a learning framework for constructing a data-driven heuristic to guide Beam Search in combinatorial optimization problems, and specifically one sequence problem that arises in bioinformatics, the *Variable Gapped Longest Common Subsequence Problem* (VGLCSP). A feed-forward neural network (FFNN) is used to parameterize the heuristic function. Instead of relying on gradient-based training, the network weights are optimized using a Genetic Algorithm (GA), in particular its popular versions called the *Random key generic algorithm* (RKGA) and a biased RKGA (BRKGA) approach.

Each individual in the GA population represents a complete set of neural network weights. The quality of an individual is evaluated by embedding the neural network inside the *Iterative Multi-source Beam Search* (IMSBS) framework (proposed in the paper []) to guide the search in the process of forward passes of the BS, and measuring the average solution quality obtained on a set of training instances. Validation instances are used exclusively to monitor generalization.

Formally, the learning problem is defined as:

$$\max_{\theta} \frac{1}{|T|} \sum_{I \in T} f(\text{IMS_BeamSearch}(I, h_\theta)),$$

where T denotes the training set, h_θ is the neural heuristic parameterized by weights θ, and $f(\cdot)$ measures solution quality.

The paper is organized as follows. TODO

2 Iterative Multi-Source Beam Search for GLCS Problem

TODO

3 Utilizing Feed-Forward Neural Networks for Guiding the Search

The heuristic is represented by a *multi-layer perceptron* with an arbitrary number of hidden layers and user-defined units per layer. Supported activation functions include `hyperbolic tangent`, `ReLU`, and `sigmoid`.

For each layer l, forward propagation is defined as:

$$\mathbf{z}^{(l+1)} = W^{(l)}\mathbf{a}^{(l)} + b^{(l)}, \quad \mathbf{a}^{(l+1)} = \phi(\mathbf{z}^{(l+1)}),$$

where $W^{(l)}$ and $b^{(l)}$ are the weight matrix and bias vector, respectively, and ϕ denotes the activation function.

All weights and biases are flattened into a single vector, which forms the chromosome of a GA individual.

3.1 Fitness Evaluation

Given a candidate weight vector:

1. The weights are loaded into the FFNN.
2. IMSBS is executed on each training instance using the neural heuristic (guiding the search in the forward iterations as more sensible than the backward iterations).
3. The size of the best sequence obtained for each instance is recorded.

The fitness value is computed as the average solution quality over all training instances. Validation performance is computed analogously on a separate dataset and is used only for reporting.

(Q0) Speed up – evaluate fitness value by using parallelization.

3.2 Feature Extraction for the Variable-Gap LCS

To guide the search process, each partial solution (node) is represented by a fixed-length numerical feature vector capturing both the current state of the construction and the structure of the remaining search space. These features are designed to be instance-independent and comparable across problem sizes.

Let $S = \{s_1, \ldots, s_k\}$ denote the set of input sequences, and let p_i be the current matched position in sequence s_i at a given node. The vector $\mathbf{p}^L = (p_1, \ldots, p_k)$ is first normalized by the corresponding sequence lengths,

$$p_i \leftarrow \frac{p_i}{|s_i|},$$

so that progress in each sequence lies in $[0, 1]$ and becomes independent of absolute input sizes.

From this normalized position vector, four statistical descriptors are extracted: maximum, minimum, mean, and standard deviation (4 features). These quantities characterize, respectively, the most advanced and least advanced sequences, the average depth of the search, and the imbalance of progress among sequences. In addition, *the current length of the partially constructed subsequence* is included as a feature (as the fifth feature). Together, these values provide a compact description of the current search state.

To capture the feasibility of future extensions, gap-related features are computed from the positions of remaining sequences (subproblem) of input sequences. For each sequence s_i, all gap bounds associated with positions strictly after the current node position are collected. Symbolically, for each v, Gap_i^{min} is calculated as $\min\{G_i(j) \mid j = p_i^L, i = 1, \ldots, m\}$, Gap_i^{max} is calculated as $\max\{G_i(j) \mid j = p_i^L, i = 1, \ldots, m\}$, Gap_i^{avg} as $\mathrm{AVG}\{G_i(j) \mid j = p_i^L, i = 1, \ldots, m\}$, and Gap_i^{std} as $\mathrm{STD}\{G_i(j) \mid j = p_i^L, i = 1, \ldots, m\}$, where $G_1, \ldots, G_m$ are gap constraints associated to $s_1, \ldots, s_m$ respectively speed up the calc. of G-stats – preprocessing... . This gives us additionally 4 features; in overall 9 features are used for the **feature configuration 1** (F1). These statistics for gap constrains summarize the remaining flexibility of the instance: large average or maximum gaps indicate greater freedom for future matches, whereas small minimum gaps and large variability typically signal tighter constraints and increased difficulty. Additional feature $|\Sigma|$, this 10 features, created the **feature configuration 2** (F2); When adding m (number of input sequences) to F2, we obtain **feature configuration 3** (F3). (Q1) uestion: s there a need for normalization of gap values of respective subproblem... (Q2) other features?

The alphabet size $|\Sigma|$ and the number of sequences (m) as the attributes provide global structural information about the instance and help the learning model adapt to varying problem scales.

Finally, all features are standardized to zero mean and unit variance prior to being passed to the learning component. The utilization of normalization is know to contribute decreased sensibility of learning process when ML models are trained. Overall, the resulting representation combines information about (i) the current construction progress, (ii) the remaining gap-constraint landscape,

and (iii) global instance properties, yielding between 9 and 11 features depending on the configuration. This compact yet expressive encoding enables effective learning-based guidance of the search for high-quality variable-gap longest common subsequences.

4 Genetic Algorithm

The optimization process follows a population-based evolutionary scheme.

4.1 Initialization

Each individual is initialized by sampling weights uniformly from a predefined interval $[-w, w]$ (the value w parametrized, but fixed to 1).

4.2 Elitism

A fixed number of top-performing individuals are copied unchanged into the next generation.

4.3 Mutation

Several individuals are generated completely at random (random immigrants). This mechanism promotes exploration and maintains population diversity.

4.4 Offspring Generation

Three alternative parent-selection strategies are supported:

- **RKGA**: Two parents are selected uniformly at random, and each gene is inherited from either parent with probability 0.5.
- **BRKGA**: One parent is selected from the elite set and one from the non-elite set. Each gene is inherited from the elite parent with a predefined probability.
- **Lexicase Selection**: Training instances are randomly shuffled. Individuals are filtered iteratively by retaining those achieving the best performance on each instance. Parents are selected randomly from the remaining candidates.

We opt to use the *BRKGA* as the first option to work well.

4.5 Replacement and Termination

The new population consists of elites, mutants, and offspring. The algorithm terminates once a predefined time limit is reached (in this case we used 2 hours). The best individual encountered during training is continuously tracked and stored. (Q3) Terminate immediately when the validation value is decreased?

5 Overall Learning Procedure

The learning process alternates between evolutionary optimization of network weights and evaluation through (iterative multi-source) Beam Search. Importantly, the neural network does not directly predict solutions; instead, it learns how to *guide* the search procedure. This places the method within the class of neuro-evolutionary hyper-heuristics.

6 Algorithm

Algorithm 1 GA-Based Learning of a Beam Search Heuristic

```
Initialize population P with random neural network weights
Evaluate all individuals on training set T
Store best solution θ*
start ← current time
while time limit T_max has not exceeded do
    Sort P by fitness (descending) // Section 2.1
    Initialize empty population P'
    Copy top n_elites individuals from P to P'
    for i = 1 to n_mutants do
        Create random individual x
        Evaluate x on T
        Add x to P'
        Update θ* if improved
    end for
    for i = 1 to n_offspring do
        Select parents using RKGA, BRKGA, or Lexicase
        Generate offspring by crossover
        Evaluate offspring on T
        Add offspring to P'
        Update θ* if improved
    end for
    P ← P'
end while
return best weights θ*
```

6.1 Explanation of the GA-Based Learning Procedure

The proposed algorithm employs a Genetic Algorithm (GA) to learn the parameters of a neural network that guides a beam search heuristic. The objective is to optimize a parameter vector θ, representing the neural network weights, such that the induced heuristic achieves high performance on a given training set T.

Initialization. The procedure starts by generating an initial population P of candidate solutions, where each individual corresponds to a randomly initialized set of neural network weights. All individuals are evaluated on the training set T, yielding a fitness value that reflects the quality of the corresponding heuristic. The best-performing individual found so far is stored as θ^*.

Evolutionary Process. The algorithm iteratively improves the population until a predefined time limit $T_{\max}$ is reached. At each iteration, the population is sorted in descending order of fitness, ensuring that higher-quality solutions are prioritized. A new population P' is then constructed using three complementary mechanisms:

- **Elitism.** A fixed number n_{elites} of top-performing individuals are directly copied from the current population P to P'. This guarantees that the best solutions are preserved across generations and prevents degradation of solution quality.
- **Mutation (Random Immigrants).** A set of $n_{mutants}$ new individuals is generated by random initialization. These individuals introduce diversity into the population and help prevent premature convergence. Each mutant is evaluated on the training set T and added to P'. If a mutant outperforms the current best solution, θ^* is updated accordingly.
- **Crossover (Recombination).** The remaining $n_{offspring}$ individuals are generated via recombination. Parent solutions are selected using a specified selection strategy, such as Random-Key Genetic Algorithm (RKGA), Biased RKGA (BRKGA), or Lexicase selection. Offspring are created by combining genetic material from the selected parents. Each offspring is evaluated and added to P', and θ^* is updated if an improved solution is found.

Population Update and Termination. After generating all individuals, the current population P is replaced by the new population P'. This evolutionary process continues until the time limit $T_{\max}$ is reached. The algorithm then returns the best solution θ^*, representing the neural network weights that yield the most effective beam search heuristic on the training set.

Key Properties. The proposed approach exhibits several desirable characteristics:

- **Exploration–Exploitation Balance:** The combination of elitism, mutation, and crossover ensures a balance between intensification around high-quality solutions and diversification across the search space.
- **Gradient-Free Optimization:** The GA operates directly on neural network parameters without requiring gradient information, making it suitable for non-differentiable or noisy objective functions.
- **Anytime Behavior:** The best solution θ^* is continuously updated, allowing the algorithm to return a high-quality solution at any point during execution.

As the best optimizer, the BRKGA setting used in training: $n_{pop} = 20, n_{elites} = 1, p_e = 0.5, p_m = 7$.

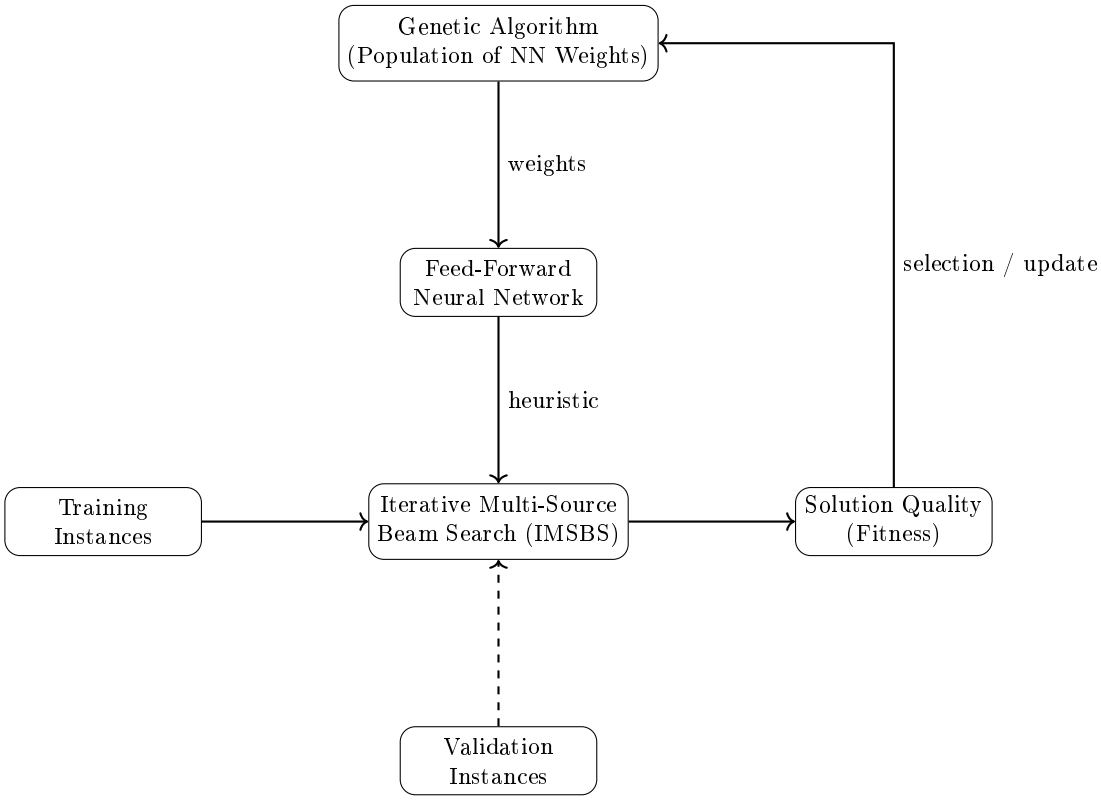

Fig. 1. Learning framework integrating a Genetic Algorithm with Iterative Multi-Source Beam Search (IMSBS). The GA evolves neural network weights. For each individual, the FFNN provides a heuristic used by IMSBS on training instances. The resulting solution quality defines fitness, which drives evolutionary selection. Validation instances are used only for monitoring generalization.

7 Discussion

The proposed framework learns heuristics rather than explicit solutions. The neural network is optimized exclusively through downstream search performance, enabling learning in non-differentiable settings. This tightly coupled integration of evolutionary learning and the classical optimization algorithm Beam Search allows the system to adapt its guidance strategy to the structure of the problem instances.

8 Preliminary results

IMSBS setting (grid search used for tuning the algorithm):

- $\beta = 1000$, $h' = h5$, $\#imbs_iters = 1000, \#num_roots = 10$

Table 1. Comparison of objective values between IMSBS and LIMSBS. Best results per instance category are in bold.

m	n	$\|\Sigma\|$	IMSBS	LIMSBS
2	50	2	**37.7**	**37.7**
2	50	4	**30.1**	29.7
2	100	2	**72.9**	72.5
2	100	4	**61.7**	60.3
2	200	2	**139.7**	139.3
2	200	4	**125.2**	120.9
2	500	2	281.9	**295.4**
2	500	4	**302.9**	285.4
3	50	2	**31.2**	**31.2**
3	50	4	**22.9**	**22.9**
3	100	2	**58.5**	**58.5**
3	100	4	**48.5**	48.0
3	200	2	**106.3**	103.4
3	200	4	**97.4**	95.6
3	500	2	133.5	**143.1**
3	500	4	**230.3**	218.8

m	n	$\|\Sigma\|$	IMSBS	LIMSBS
5	50	2	**20.5**	**20.5**
5	50	4	**15.3**	**15.3**
5	100	2	**31.5**	31.0
5	100	4	25.9	**26.9**
5	200	2	35.2	**41.6**
5	200	4	34.0	**37.2**
5	500	2	31.0	**39.3**
5	500	4	36.3	**46.2**
10	50	2	**11.4**	11.2
10	50	4	**7.1**	**7.1**
10	100	2	10.5	**12.0**
10	100	4	7.6	**8.9**
10	200	2	11.8	**12.2**
10	200	4	7.6	**8.2**
10	500	2	12.5	**12.9**
10	500	4	7.5	**8.3**

Learning IMSBS setting uses the same setting, to be completely fair with the comparison. The neural network configuration in this case, obtained after running several 2-layer (n_1, n_2) and 3-layer configurations (n_1, n_2, n_3) for $n_1, n_2, n_3 \in \{5, 10\}$ (neurons) and $n_1 \geq n_2 \geq n_3$. We set up the same activation function in all layers choosing among the three (most) known $\{\texttt{tanh}, \texttt{ReLU}, \texttt{sigmoid}\}$. The best one uses `ReLU` at each layer, and besides instance-based features, it additionally incorporate the size of alphabet Σ (F2). The configurations of NNs are obtained only after 2 hours of training is a 3-layer perception with 10 neurons each.

9 Conclusions

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